Topic IX: Computer Operations and Programming

Chapter

41. Accuracy Standards

41 Accuracy Standards

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1. ACCURACY VERSUS PRECISION IN STANDARDS

The distinction between accuracy and precision is an important concept in standards for surveying and mapping. As discussed in Chap. 11, *precision* is the measure of consistency between observations, while *accuracy* is the measure of nearness to the true value. In surveying, a "true" value is usually not known, as no measurement is absolutely perfect. Instead, a measurement from a higher-order survey is typically used to evaluate the accuracy of a measurement.

Many surveying standards, such as those requiring redundancy of measurements or closure of figures, are really standards for precision, since they do not provide for comparison with an accepted value. For example, while standards establishing minimum closure ratios for traverses may check the precision of the points in the traverse, the points may be inaccurate if a scale error exists in the electronic distance measurement (EDM) used.

2. NATIONAL MAP ACCURACY STANDARDS

The *United States National Map Accuracy Standards* are possibly the oldest standards for comparing the relative accuracy of published maps. These standards have been superseded by the *Federal Geographic Data Committee (FGDC) Geospatial Positioning Accuracy Standards* of 1998, but are included for historical interest.

The *U.S. National Map Accuracy Standards* were originally developed by a committee established by the American Society for Photogrammetry and Remote Sensing in 1937. That draft inspired several U.S. agencies to begin studies of standards for map accuracy. The final standards were formally issued by the U.S. Bureau of the Budget in 1941, and were revised in 1947. The standards were intended for application by all federal agencies producing maps.

The standards require that the accuracy of a map be tested by comparison of well-defined points, such as the intersection of two roads, whose location or elevation is shown on the map, with positions determined by surveys of a higher accuracy. Generally, to test a map, 20 or more well-defined points are used.

For testing of horizontal accuracy, the standards require that for maps with scales larger than 1:20,000, not more than 10% of the points tested can be in error by more than 1/30 in. as measured on the publication scale. For maps with scales of 1:20,000 or smaller, not more than 10% of the points tested can be in error by more than 1/50 in. For testing of vertical accuracy (with contours), not more than 10% of the elevations tested can be in error by more than one-half the contour interval.

As an example of the *U.S. National Map Accuracy Standards* for horizontal accuracy, for a 1:24,000 scale topographic map like those produced by the United States Geological Survey (USGS), not more than 10% percent of the points tested can be in error by more than 1/50 in. At that scale, an error of 1 in represents an actual error of

$$1 \text{ in} = \frac{24{,}000 \text{ in}}{12 \ \frac{\text{in}}{\text{ft}}} = 2000 \text{ ft}$$

Therefore, 1/50 in represents 40 ft on the maps, and that value would be the maximum horizontal error allowable under the standards for tested points.

As an example of application of that standard for vertical accuracy, for a map with a 10 ft contour interval, no more than 10% of tested points can have an error greater than 5 ft.

3. FEMA ACCURACY STANDARDS

The Federal Emergency Management Administration (FEMA) has promulgated accuracy standards for light detection and ranging (LiDAR) data used in the agency's flood plain management maps. Those standards address system calibration, flight planning, acceptable control points, and post processing. In addition, the FEMA standards address quality control and quality assurance (QC/QA). As opposed to the requirements for selection of features that can be identified on a map under the *U.S. National Map Accuracy Standards*, the test points for these standards may be selected

at random since, with LiDAR mapping, a horizontal position is associated with an elevation for each measured point.

The standards regarding QC/QA require that the elevation data have a maximum *root-mean-squared error* (RMSE) of 15 cm. As used in the FEMA accuracy standards, RMSE is the square root of the average of the squared differences between the measured elevations for selected points, compared to the *check elevations*, elevations for the same points measured by an independent survey of higher accuracy. (See Chap. 11 for further discussion of the calculation of the RMSE.)

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x_i')^2}{n}} \qquad 41.1$$

Based on random error theory, if there is a normal distribution of errors, 95% of the errors in a sufficiently large sample of measurements should be less than 1.96 times the RMSE. This value (1.96 times the RMSE) is often called the *2 sigma error*.

The standards require that at least 20 points be sampled for each of the main categories of ground cover in the study area, allowing for one point to fail the test without compromising the 95% confidence level. Examples of categories of ground cover provided in the standards include:

- bare earth and low grass (e.g., plowed fields, lawns, golf courses)
- high grass and crops (e.g., hay fields, corn fields, wheat fields)
- brush lands and low trees (e.g., chaparrals, mesquite, mangrove swamps)
- ground fully covered by trees (e.g., hardwoods, evergreens, mixed forests)
- urban areas (e.g., high, dense manmade structures)
- sawgrass

Since the criterion for measurement accuracy is based on the assumption of a normal distribution of errors, these standards require examination of the data for possible skew, which would be suggestive of systematic error. If the errors are absolutely random, the mean of the errors is statistically likely to be zero. The FEMA standards consider a mean of the errors greater than ±2 cm as an indicator of unacceptable skew.

Example 41.1

The table shown gives a series of elevations obtained by means of an airborne LiDAR survey. The check elevations were obtained from a real-time kinematic (RTK) GPS survey of a brush lands area. What is the RMSE for the data?

n	latitude	longitude	LiDAR, x_i (m)	GPS, x_i' (m)	$x_i - x_i'$ (m)	$(x_i - x_i')^2$ (m²)
1	9°55′10.25″ N	85°20′12.45″ W	17.43	17.47	−0.03	0.001
2	9°55′13.22″ N	85°20′30.23″ W	19.93	19.87	0.06	0.004
3	9°55′17.31″ N	85°20′50.22″ W	18.04	18.32	−0.28	0.078
4	9°55′21.67″ N	85°21′12.67″ W	21.67	21.67	0.00	0.000
5	9°55′25.50″ N	85°21′29.43″ W	19.08	19.05	0.03	0.001
6	9°55′28.41″ N	85°21′09.76″ W	18.87	19.05	−0.18	0.033
7	9°55′31.67″ N	85°20′49.43″ W	18.29	18.59	−0.30	0.093
8	9°55′34.93″ N	85°20′41.19″ W	17.40	17.47	−0.07	0.005
9	9°55′37.13″ N	85°20′31.84″ W	16.55	16.52	0.03	0.001
10	9°55′40.85″ N	85°20′11.32″ W	18.17	18.17	0.00	0.000
11	9°55′43.32″ N	85°20′26.56″ W	19.48	19.42	0.06	0.004
12	9°55′46.64″ N	85°20′37.21″ W	19.02	19.11	−0.09	0.008
13	9°55′49.25″ N	85°20′50.78″ W	19.96	19.84	0.12	0.014
14	9°55′52.33″ N	85°21′10.62″ W	20.70	20.67	0.03	0.001
15	9°55′55.27″ N	85°21′28.42″ W	20.79	20.82	−0.03	0.001
16	9°55′58.21″ N	85°21′14.91″ W	18.50	18.56	−0.06	0.004
17	9°56′00.81″ N	85°20′55.52″ W	20.51	20.45	0.06	0.004
18	9°56′04.54″ N	85°20′40.15″ W	21.79	21.92	−0.13	0.017
19	9°56′07.76″ N	85°20′28.68″ W	23.84	23.77	0.07	0.005
20	9°56′09.21″ N	85°20′12.01″ W	24.99	24.72	0.27	0.073
					$\bar{x}$ = −0.02	Σ = 0.347

Solution

The RMSE for the data is

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x_i')^2}{n}} = \sqrt{\frac{0.347}{20}} = 0.13 \text{ m} \quad (13 \text{ cm})$$

The results show a maximum RMSE of less than 15 cm and a mean error of 0.02 meters (2 cm), so the survey appears to meet the FEMA accuracy standards.

4. FGDC ACCURACY STANDARDS

The Federal Geographic Data Committee (FGDC) is a U.S. government committee that promotes the coordination, use, sharing, and dissemination of geospatial data on a national basis. The committee is chaired by a person nominated by the Department of the Interior, with a nominee from the Office of Management and Budget as vice chair. Other members are representatives from various federal government agencies, including the Departments of Agriculture, Commerce, Defense, Energy, Housing and Urban Development, Interior, State, and Transportation; the Environmental Protection Agency; FEMA; the Library of Congress; the National Aeronautics and Space Administration; the National Archives and Records Administration; and the Tennessee Valley Authority. The FGDC has developed comprehensive geospatial positioning accuracy standards, known as the *FGDC Geospatial Positioning Accuracy Standards*.

Part 2: Standards for Geodetic Networks

Part 2 of the *Geospatial Positioning Accuracy Standards* applies to geodetic networks. The standards provide a common method for classifying the accuracy of geodetic control points, which are usually part of a control network for surveying and mapping. Such geodetic surveys are the basis for other surveys and must be performed to more rigorous standards than those performed for general engineering, construction, or topographic mapping purposes. These standards supersede the *Standards and Specifications for Geodetic Control Networks* promulgated by the Federal Geodetic Control Commission in 1984.

For geodetic control points, accuracy under the FGDC standards is determined for each geodetic component (i.e., horizontal control, ellipsoidal height, and orthometric height). The accuracy standards for geodetic networks are not measured by closure within a survey, but by relationships with the network datum, taking into account crustal motion, refraction, and any other systematic effects known to influence survey measurements.

Survey accuracy is checked by comparing the survey results against established controls using a minimally constrained least-squares adjustment for the 95% confidence level. This accuracy is reported as meeting a level of precision of 1, 2, or 5 mm; 1, 2, or 5 cm; or 1, 2, 5, or 10 m.

Part 3: National Standard for Spatial Data Accuracy

Part 3 of the *Geospatial Positioning Accuracy Standards* provides a methodology for estimating the positional accuracy of points on maps and in digital geospatial data, with respect to georeferenced control points of higher accuracy.

These standards are similar to the FEMA standards, in that they use the RMSE to estimate positional accuracy, comparing map values with measurements based on an independent source of higher accuracy for identical points. The standards require that a minimum of 20 check points, distributed to reflect the geographic area of interest, be tested. As with the FEMA standards, when 20 points are tested, the 95% confidence level allows one point to fail the threshold specifications.

Part 4: Architecture, Engineering, Construction, and Facilities Management

Part 4 of the *Geospatial Positioning Accuracy Standards* provides accuracy standards for engineering drawings, maps, and surveys for planning, design, construction, operation, maintenance, and management of facilities, installations, structures, transportation systems, and related projects. This includes related geographical information systems (GIS), computer-aided drafting and design (CADD), and automated mapping/facility management (AM/FM) products.

The standards require that such projects be referenced to local boundary control commonly used in the project area. Where practical and feasible, horizontal coordinates should be referenced to national coordinate systems, such as the State Plane Coordinate System based on the North American Datum of 1983 (NAD 83) and vertical coordinates based on the North American Vertical Datum of 1988 (NAVD 88). For the control surveys for such projects, two types of survey accuracies may be specified: positional accuracy and relative closure ratio accuracy.

Positional accuracy is the difference between the results of a measurement of a location of a point and the true position of that point. When positional accuracy is used to determine survey accuracy, the accuracy is checked in the manner required for geodetic control surveys, as described in Part 1 of the FGDC standards. The survey results are compared to established controls using a minimally constrained least squares adjustment for the 95% confidence level, consistent with the engineering or construction applications or specifications. Generally, horizontal and vertical control point accuracies should be twice as accurate as tolerances required for features or objects on the resulting maps.

Relative closure ratio accuracy is the linear misclosure, such as is determined by a closed traverse, divided by the total length of the traverse. Relative closure ratio accuracy is often the prevailing standard when certain closure ratios are specified in state codes and/or state minimum technical standards. No simple correlation between relative closure accuracies and 95% radial positional accuracies exists. Therefore, knowing the closure is not sufficient to determine whether a survey meets a specified feature accuracy requirement. As a result, positional accuracy standards should be used instead of closure accuracy standards where practical and allowable. Relative closure ratio accuracy could be considered a measure of precision, rather than accuracy, but is included to allow use of surveys meeting various state standards.

For the resulting maps of this type, the survey accuracy is checked in the manner required for maps under Part 2 of the FGDC standards by testing both horizontal positions of planimetric features and elevations against measurements from a source of higher accuracy and determining the RMSE.

Part 5: Standards for Nautical Charting Hydrographic Surveys

Hydrographic surveys, as covered by Part 5 of the *Geospatial Positioning Accuracy Standards*, are defined as surveys conducted to determine the configurations of the bottoms of water bodies, and to identify and locate all natural and man-made features that may affect navigation. Defining the accuracy of such surveys is complex, due to the fact that several components are involved. These include the accuracy of the horizontal positioning of each sounding; the accuracy of the

determination of the vertical datum used in the survey; the accuracy of any corrections that are applied to the soundings, such as tide, draft, settlement, and squat corrections; and the accuracy of the soundings themselves. As a result, several considerably different approaches are used for defining the accuracy of hydrographic surveys.

Typically, both the horizontal position of each sounding and the vertical measurement at that position are indirect measurements determined by mathematical relationships to direct measurements. For example, horizontal positions may be derived from direct angle and distance measurements establishing control points on the shore, together with direct GPS measurements on those control points and on the sounding positions. Similarly, the vertical measurement (depth) may be derived from direct tide gauge measurements establishing the vertical datum for the survey and for tide corrections, direct fathometer measurements, and direct measurement of other corrections for the soundings.

As a result of the nature of measurements associated with hydrographic surveys, the FGDC standards require that all contributions to the total error be propagated (combined) using a statistical method. One method used for this purpose for less complex relationships is that of using the square root of the sum of the squares of all of the contributing errors. For example, if three direct components of an indirect measurement have errors of 0.01, 0.02, and 0.03 ft, the total error combined by this method would be

$$(0.01\ \text{ft})^2 + (0.02\ \text{ft})^2 + (0.03\ \text{ft})^2 = \sqrt{0.0014\ \text{ft}} = 0.04\ \text{ft}$$

The propagation of errors for hydrographic surveys is especially complex with the increased use of multi-beam fathometers. With such equipment, roll, pitch, heading, refraction, and other errors associated with both the horizontal and vertical positions of each sounding becomes critical. Formal procedures for calculation of the total propagated error (TPE) for multi-beam hydrographic surveys have been developed, but these procedures are beyond the scope of this chapter. For horizontal errors, those procedures take into account refraction errors; roll, pitch, and heading errors; positioning system errors; sensor coordinate offset errors; transducer head misalignment errors; and latency errors between sensors. For vertical errors, the calculation of TPE takes into account sounder measurement error; sound speed correction errors; dynamic draft measurement errors; heave measurement errors; tide or water-level errors; roll, pitch, and heading errors; and transducer head misalignment errors.

Classifications of Hydrographic Surveys

To accommodate different accuracy requirements in a systematic manner, the FGDC standards define four orders of survey, as shown in Table 41.1.

Table 41.1 *Minimum Standards for Hydrographic Surveys*

classification	special order	first order	second order	third order
typical application	harbors, berths, & critical depth areas	channels & coastal areas ≤ 100 m depth	≤ 200 m depth	offshore > 200 m depth
horizontal accuracy	2 m	5 m + 5% of depth	20 m + 5% of depth	150 m + 5% of depth
depth accuracy	$a = 0.25$ m $b = 0.0075$	$a = 0.5$ m $b = 0.013$	$a = 1.0$ m $b = 0.023$	$a = 2.0$ m $b = 0.046$
maximum line spacing	100% coverage required	greater of 25 m or 3 × average depth, 100% coverage in some places	greater of 200 m or 4 × average depth	4 × average depth

Note: Accuracies refer to accuracy at the 95% confidence level in reference to the datum reference frame.

Datums for Hydrographic Surveys

The *Geospatial Positioning Accuracy Standards* require that the North American Datum of 1983 (NAD 83) be used as the horizontal datum for hydrographic surveys and that depths be referenced to the applicable chart datum, rather than geodetic vertical datums such as the North American Vertical Datum of 1988 (NAVD 88) or the National Geodetic Vertical Datum of 1929 (NGVD 29). The Mean Lower Low Water (MLLW) datum is typically used for Atlantic, Pacific, and Gulf coast charts, and the International Great Lakes Datum (1985) is typically used for the Great Lakes. Other water level-based datums are used for lakes and rivers.

Horizontal Accuracy

The *Geospatial Positioning Accuracy Standards* require primary shore control points to be located by ground survey methods to a relative accuracy of 1 part in 100,000. When such points are established by satellite positioning, the error should not exceed 10 cm at the 95% confidence level. Secondary shore control points (those not used for extending the control) should have errors not exceeding 1 part in 10,000 for ground survey techniques or 50 cm using satellite positioning.

Under these standards, *horizontal spatial accuracy* is defined as the two-dimensional circular error of horizontal coordinates at the 95% confidence level. Errors in latitude and longitude can be combined to create a one-dimensional radial error, such as a circle of equal probability.

Vertical Accuracy

In determining the depth accuracy of the reduced depths, the sources of individual errors should be quantified and statistically combined to obtain a TPE at the 95% confidence level. As an additional check on data quality, redundant depths from crossline intersections should be measured.

For wrecks and obstructions with less than 40 m clearance, the least depth over them should be determined by high definition sonar examination or physical examination (i.e., diving). Mechanical sweeping may be used when guaranteeing a minimum safe clearance depth.

Due to the rapidly increasing size of commercial vessels, there is a demand for more accurate and reliable knowledge of the water depths, in order to safely exploit the maximum cargo capabilities. Therefore, depth accuracy standards in critical areas are more stringent than those established in the past.

Tidal Observation Accuracy

Unless 3-D positioning is used, which would eliminate the need for tide corrections, tidal height observations should be made throughout the course of a survey, and for a sufficient period to provide tidal reductions for soundings and data for tidal analysis. The standards require sufficient observations be made to ensure that the total measurement error at the tide gauge, including timing errors, does not exceed ±5 cm at 95% for special order surveys or ±10 cm for other classes of surveys.

Bottom Sampling

The seabed should be sampled to determine the nature of (rocky, sandy, etc.). The nature of the seabed may also be inferred from other sensors, up to the depth required by local anchoring or trawling conditions. Normally, sampling is not required for depths greater than 200 m. Bottom sample spacing should normally be 10 times that of the main scheme line spacing. In anchorages, density of sampling should be increased.

Soundings

Sounding systems are required to be able to detect cubic features greater than 1 m in any dimension for special order surveys or first- and second-order surveys in water with depths less than 40 m. In addition to the requirements for line spacing in Table 41.1, full bottom coverage is required in selected areas where the bottom characteristics and the risk of obstructions are potentially hazardous to vessels. Crosslines (and other redundant lines) intersecting the principal sounding lines are required to confirm the accuracy of positioning, sounding, and tidal reductions. The interval between crosslines should normally be no more than 15 times that between the principal sounding lines.

5. ALTA/ACSM ACCURACY STANDARDS FOR LAND TITLE SURVEYS

Although primarily directed at the legal aspects of surveying, another set of accuracy standards are given in the standards for land title surveys developed by the American Land Title Association (ALTA) and the American Congress on Surveying and Mapping (ACSM). These standards include a requirement for land title surveys to have a relative positional accuracy of 0.07 feet (or 20 mm) + 50 ppm. Those standards define relative positional accuracy as the uncertainty due to random errors in measurements in the location of any point on the same survey at the 95 percent confidence level.

Under the ALTA and ACSM standards, relative positional accuracy may be tested by comparing the relative location of surveyed points, as measured by an independent survey of higher accuracy, and calculating the uncertainty as detailed in the FEMA standards; or by use of a minimally constrained, correctly weighted least-squares adjustment of the survey.

6. PRACTICE PROBLEMS

1. Under the Federal Geographic Data Committee accuracy standards, which vertical datum should be used for hydrographic surveys in the Gulf of Mexico?

(A) NGVD 29

(B) NAVD 88

(C) MLLW

(D) any of the above

2. The depth accuracy required for a first-order hydrographic survey in waters with a 20 m average depth is most nearly

(A) 0.01 m

(B) 0.1 m

(C) 0.6 m

(D) 1.0 m

3. What is the maximum line spacing allowable under the Federal Geographic Data Committee accuracy standards for a first-order hydrographic survey in waters with a 20 m average depth?

(A) 25 m

(B) 50 m

(C) 60 m

(D) 100 m

4. Which of the following criteria is the accuracy standard used to determine if boundary surveys meet the American Land Title Association and American Congress on Surveying and Mapping standards?

(A) standard error

(B) root-mean-squared error

(C) standard deviation

(D) relative positional accuracy

5. The table shown provides elevations for a series of bare earth points in a pine forest. The elevations are based on a LiDAR survey and check elevations for the same points measured by a real-time kinematic (RTK) global positioning system.

n	elevation from LiDAR (ft)	elevation from GPS (ft)	n	elevation from LiDAR (ft)	elevation from GPS (ft)
1	118.5	119.1	11	113.3	113.4
2	119.4	119.3	12	117.7	117.6
3	121.0	121.0	13	111.2	111.2
4	116.5	115.9	14	111.7	111.6
5	116.1	116.2	15	110.6	110.9
6	117.9	117.7	16	107.0	106.9
7	120.9	121.0	17	103.1	103.1
8	115.3	115.4	18	115.1	115.4
9	115.0	114.8	19	117.7	117.5
10	113.2	113.3	20	118.9	118.8

The root-mean-squared error for this test of the accuracy is most nearly

(A) 0.08 ft

(B) 0.11 ft

(C) 0.16 ft

(D) 0.24 ft

6. The 2 sigma error for a survey with a root-mean-squared error of 0.08 ft is most nearly

(A) 0.08 ft

(B) 0.11 ft

(C) 0.16 ft

(D) 0.24 ft

7. Which accuracy standards control for the creation of a digital elevation model (DEM)?

(A) National Map Accuracy Standards

(B) FGDC standards

(C) FGDC AM/FM standards

(D) ALTA/ACSM standards

SOLUTIONS

1. Under the Federal Geographic Data Committee accuracy standards, the local Mean Lower Low Water (MLLW) datum should be used as a vertical datum for hydrographic surveys in the Gulf of Mexico, as well as in the Atlantic and Pacific Oceans.

The answer is (C).

2. Under the Federal Geographic Data Committee accuracy standards, for a first-order hydrographic survey, the tabulated factors are

$$a = 0.5 \text{ m}$$
$$b = 0.013$$

The required depth accuracy for an average depth of 20 m is

$$\sqrt{a^2 + (bd)^2} = \sqrt{(0.5 \text{ m})^2 + ((0.013)(20 \text{ m}))^2}$$
$$= 0.56 \text{ m} \qquad (0.6 \text{ m})$$

The answer is (C).

3. The maximum line spacing allowable under the Federal Geographic Data Committee accuracy standards for the first-order hydrographic survey in waters with a 20 m average depth is the greater of 25 m or 3 times the average depth. With an average depth of 20 m, the maximum line spacing would be (3)(20 m) = 60 m.

The answer is (C).

4. The American Land Title Association and American Congress on Surveying and Mapping standards use the relative positional accuracy as derived from a minimally constrained, correctly weighted least-squares adjustment of the survey to determine if boundary surveys meet the standards.

The answer is (D).

5. Find the difference between the elevation and the check elevation and the square of the difference between those elevations for each point surveyed, then sum those values to find $\bar{x}$ and $\sum$.

n	LiDAR, x_i (ft)	GPS, x_i' (ft)	$x_i - x_i'$ (ft)	$(x_i - x_i')^2$ (ft)
1	118.5	119.1	−0.6	0.4
2	119.4	119.3	0.1	0.0
3	121.0	121.0	0.0	0.0
4	116.5	115.9	0.6	0.4
5	116.1	116.2	−0.1	0.0
6	117.9	117.7	0.2	0.0
7	120.9	121.0	−0.1	0.0
8	115.3	115.4	−0.1	0.0
9	115.0	114.8	0.2	0.0
10	113.2	113.3	−0.1	0.0
11	113.3	113.4	−0.1	0.0
12	117.7	117.6	0.1	0.0
13	111.2	111.2	0.0	0.0
14	111.7	111.6	0.1	0.0
15	110.6	110.9	−0.3	0.1
16	107.0	106.9	0.1	0.0
17	103.1	103.3	−0.2	0.0
18	115.1	115.4	−0.3	0.1
19	117.7	117.5	0.2	0.0
20	118.9	118.8	0.1	0.0
			$\overline{x} = 0.0$	$\sum = 1.0$

The root-mean-squared error is

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x_i')^2}{n}} = \sqrt{\frac{1.0}{20}} = 0.22 \text{ ft} \quad (0.24 \text{ ft})$$

The answer is (D).

6. The 2 sigma error for a survey is 1.96 times the root-mean-squared error (RMSE), so the 2 sigma error for a survey with an RMSE of 0.08 ft is

$$(1.96)(0.08 \text{ ft}) = 0.1568 \text{ ft} \quad (0.16 \text{ ft})$$

95% of the errors should be less than 0.16 ft.

The answer is (C).

7. The Federal Geographic Data Committee (FGDC) standards control for the creation of a digital elevation model.

The answer is (B).